

Milestone 02: Independent Research Plan

Part 01: Research Question

My research question is: "How do patients' wait times and procedure volumes for CT scans compare to MRI scans across different quarters of the year, and are there identifiable seasonal trends in radiology throughput?" I want to explore whether certain times of the year see a significant increase in specific imaging modalities, which can impact clinical workflow and patient care.

Part 02: Data Sources

I will retrieve data from a public healthcare repository, specifically using datasets from CMS.gov (Hospital Outpatient Quality Reporting) or a secondary healthcare dataset from Kaggle. I plan to use Python's panda's library to import the data into a data frame, clean the entries, and perform descriptive statistics to answer my research question.

Part 03: Final Deliverable Description

My final delivery will consist of a Jupyter Notebook containing the Python script used for the analysis. My notebook will include data visualizations, specifically a bar chart to compare wait times between modalities and a line graph to illustrate volume trends over time. Also, I will provide a README.md file on GitHub that outlines the methodology, the libraries used, such as: Matplotlib or Seaborn, and a summary of the findings.